

Chapter 2G

Interstitial Lung Disease, Occupational Lung Disease, and Sarcoidosis

Original pulmonary medicine chapter for graph-RAG oriented study: mechanisms, diagnostic anchors, illness scripts, management pathways, and concept relationships.

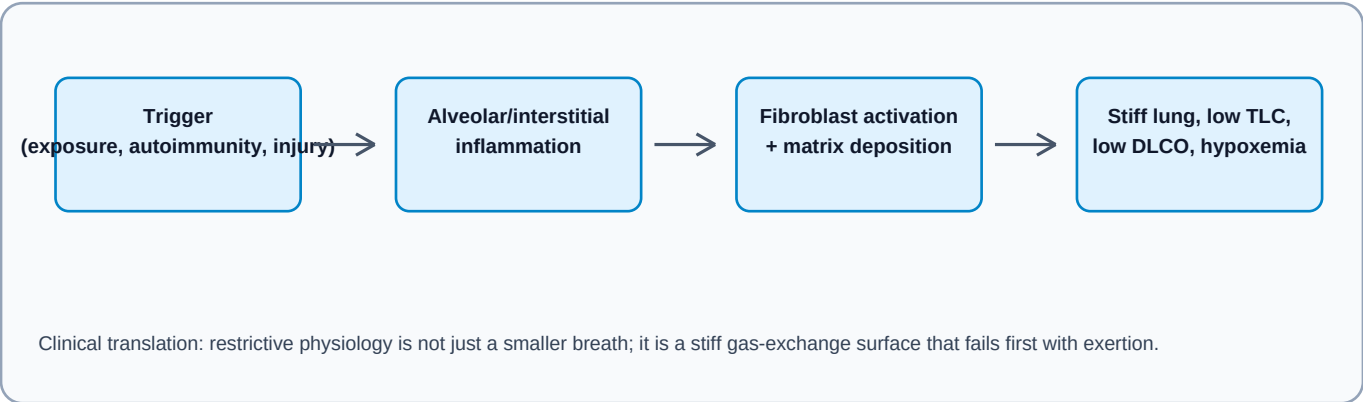
How to use this chapter

This segment is designed to connect symptoms, physiology, imaging patterns, exposures, immune mechanisms, complications, and management choices. It intentionally uses repeated numeric anchors and named relationships so a graph RAG system can extract nodes such as **restrictive physiology**, **usual interstitial pneumonia**, **asbestos exposure**, **noncaseating granuloma**, and **pulmonary hypertension** with meaningful edges.

1. Why this matters clinically

Interstitial lung diseases are a large group of disorders in which the normally thin alveolar-capillary interface becomes inflamed, scarred, granulomatous, or filled with abnormal material. The patient often does not present saying, 'I have fibrosis.' The presentation is more subtle: progressive exertional dyspnea, dry cough, oxygen desaturation while walking, fine inspiratory crackles, or an abnormal chest CT performed for another reason. These diseases matter in internal medicine because they are frequently misread as asthma, COPD, pneumonia, heart failure, deconditioning, or anxiety before the restrictive pattern is recognized.

The central physiologic problem is loss of compliant gas-exchanging lung surface. In obstructive disease, patients struggle to empty air. In many interstitial diseases, patients struggle to expand stiff lungs and to move oxygen across a thickened or damaged membrane. This produces a pattern of reduced total lung capacity, reduced forced vital capacity, preserved or increased FEV1/FVC ratio, and often reduced diffusion capacity for carbon monoxide. The DLCO is especially useful because it falls when the alveolar-capillary membrane is damaged or when pulmonary vascular surface area is reduced.



2. Core illness scripts

Illness script	Typical patient story	Physiology	Imaging clue	Management anchor
Idiopathic pulmonary fibrosis	Age usually >50 years; progressive exertional dyspnea, dry cough, bibasilar crackles, possible clubbing; no clear exposure or systemic autoimmune disease.	Restrictive PFTs with reduced FVC and reduced DLCO; exertional hypoxemia may precede resting hypoxemia.	High-resolution CT with basal/subpleural reticulation, traction bronchiectasis, and honeycombing when usual interstitial pneumonia is present.	Confirm pattern, exclude mimics, consider antifibrotic therapy, oxygen for hypoxemia, pulmonary rehab, transplant referral when appropriate.
Connective tissue disease ILD	Dyspnea plus Raynaud phenomenon, inflammatory arthritis, mechanic's hands, sicca symptoms, proximal weakness, rash, or known systemic sclerosis, rheumatoid arthritis, myositis, lupus, or Sjogren disease.	Restrictive pattern and reduced DLCO; pulmonary hypertension may coexist, especially in systemic sclerosis.	NSIP pattern is common in several autoimmune ILDs; UIP can occur in rheumatoid arthritis.	Treat the systemic disease and lung phenotype; involve rheumatology and pulmonology early.
Hypersensitivity pneumonitis	Recurrent cough, dyspnea, feverish episodes, or chronic fibrosis linked to birds, mold, farming, humidifiers, hot tubs, or occupational antigens.	May be restrictive, obstructive, or mixed; DLCO frequently reduced; BAL lymphocytosis can support diagnosis.	Ground glass, centrilobular nodules, mosaic attenuation, and air trapping; chronic disease may fibrose.	Identify and remove antigen; corticosteroids may help inflammatory disease; chronic fibrotic disease may behave like progressive pulmonary fibrosis.
Sarcoidosis	Young or middle-aged adult with bilateral hilar lymphadenopathy, cough, dyspnea, fatigue, uveitis, erythema nodosum, hypercalcemia, or conduction symptoms.	PFTs can be normal, restrictive, obstructive, or mixed; DLCO may decline.	Bilateral hilar adenopathy with or without parenchymal infiltrates; advanced disease may fibrose.	Diagnose with compatible picture, granulomas, and exclusion of mimics. Observe mild disease; treat organ-threatening or symptomatic disease.

3. Pathophysiology and pulmonary function pattern

Restrictive lung disease is defined physiologically by a reduced total lung capacity. Spirometry alone can suggest restriction when FVC is low and the FEV1/FVC ratio is normal or high, but lung volumes are needed to confirm true restriction. A patient with poor effort, severe obstruction with air trapping, neuromuscular weakness, obesity, or pleural disease may have a low FVC without primary interstitial fibrosis. The graph-RAG relationship is: **low FVC** suggests a volume problem, but **low TLC** confirms restriction.

DLCO adds a gas-exchange axis. DLCO is commonly decreased in interstitial lung disease, pulmonary vascular disease, emphysema, anemia, and some advanced heart failure states. DLCO may be normal in extrapulmonary restriction, such as obesity, chest wall disease, or neuromuscular weakness, because the alveolar membrane itself may be preserved. Therefore, the combination of low TLC plus low DLCO points toward parenchymal or pulmonary vascular disease, whereas low TLC with relatively preserved DLCO points toward extrapulmonary mechanical limitation.

Pattern	TLC	FVC	FEV1/FVC	DLCO	Interpretation
Classic ILD	Low	Low	Normal or high	Low	Stiff, damaged gas-exchange surface.
Extrapulmonary restriction	Low	Low	Normal or high	Normal or mildly low	Obesity, neuromuscular weakness, chest wall disease, pleural restriction.
Obstruction	Normal or high	Normal or low	Low	Variable	Asthma, COPD, bronchiectasis; air trapping may lower FVC.
Pulmonary vascular disease	Often normal	Often normal	Normal	Low	Reduced vascular bed; consider pulmonary hypertension or chronic thromboembolic disease.

High-yield numeric anchors

ILD often produces **normal/high FEV1/FVC** despite dyspnea. A decline in FVC over time is clinically important; many ILD clinics track FVC percent predicted, DLCO percent predicted, oxygen saturation at rest and with exertion, 6-minute walk distance, and need for supplemental oxygen. Progressive pulmonary fibrosis is generally suspected when symptoms, physiology, or imaging worsen over about **1 year** despite appropriate management.

4. Diagnostic approach

The diagnostic approach begins with time course and pattern recognition. Acute days-to-weeks dyspnea with diffuse opacities raises infection, pulmonary edema, diffuse alveolar hemorrhage, acute eosinophilic pneumonia, drug toxicity, or acute exacerbation of underlying ILD. Subacute weeks-to-months dyspnea suggests hypersensitivity pneumonitis, organizing pneumonia, drug-induced lung disease, sarcoidosis, malignancy-related processes, or evolving autoimmune disease. Chronic months-to-years dyspnea suggests IPF, occupational pneumoconiosis, chronic HP, sarcoidosis, CTD-ILD, or slowly progressive fibrotic disease.

History is not a formality in ILD; it is often the diagnostic test that tells you which CT pattern matters. Ask about birds, feather bedding, mold, hot tubs, farming, compost, woodworking, metalworking, stone cutting, mining, shipyard work, brake work, military or construction exposures, medications, radiation, chemotherapy, immune checkpoint inhibitors, nitrofurantoin, amiodarone, methotrexate, and biologic agents. Also ask about autoimmune clues: Raynaud phenomenon, skin thickening, inflammatory arthritis, dry eyes, dry mouth, dysphagia, proximal muscle weakness, mechanic's hands, rashes, oral ulcers, and pleuritic symptoms.

- Initial tests often include chest radiograph, high-resolution CT chest, spirometry with lung volumes, DLCO, resting and exertional oximetry, and basic labs tailored to the scenario.
- High-resolution CT is the key imaging test because it can show reticulation, honeycombing, traction bronchiectasis, nodules, ground-glass opacity, mosaic attenuation, air trapping, pleural plaques, and adenopathy.
- Serologic testing is selected when CTD-ILD is possible; examples include ANA, rheumatoid factor, anti-CCP, myositis antibodies, SSA/SSB, Scl-70, anticentromere, and others based on the clinical picture.
- Bronchoscopy may help exclude infection or malignancy, evaluate alveolar hemorrhage, or support hypersensitivity pneumonitis with BAL lymphocytosis. Surgical or cryobiopsy is reserved for cases where noninvasive data do not provide a confident diagnosis and the result would change management.

5. Idiopathic pulmonary fibrosis and progressive fibrotic ILD

Idiopathic pulmonary fibrosis is the prototype of chronic fibrosing ILD. It is idiopathic because no specific exposure, drug, connective tissue disease, or alternative cause explains it. The typical patient is an older adult with slowly progressive exertional dyspnea and nonproductive cough. Fine inspiratory crackles at the lung bases are often described as Velcro-like. Clubbing can occur but is not required. Wheezing is not the dominant finding, and repeated bronchodilator treatment without response should prompt reconsideration of the diagnosis.

The pathology and imaging pattern classically associated with IPF is usual interstitial pneumonia. UIP is a spatially heterogeneous fibrotic pattern: some lung regions look relatively preserved while adjacent regions are scarred. On HRCT, the classic UIP pattern has basal and subpleural predominance, reticulation, traction bronchiectasis, and honeycombing. Honeycombing refers to stacked cystic air spaces representing end-stage fibrotic remodeling. Ground-glass opacity may be present, but if it is the dominant feature, alternative diagnoses such as infection, edema, inflammatory ILD, HP, or organizing pneumonia become more likely.

Management is not simply giving steroids. In IPF, chronic immunosuppression can be harmful, and modern management emphasizes accurate diagnosis, antifibrotic therapy when appropriate, vaccination, smoking cessation, oxygen assessment, pulmonary rehabilitation, treatment of comorbidities such as reflux or sleep apnea when clinically relevant, palliative symptom support, and early transplant referral for eligible patients with progressive disease. Nintedanib and pirfenidone are antifibrotic agents used in IPF; nintedanib also has a role in some progressive pulmonary fibrosis phenotypes. These therapies slow decline rather than reverse established honeycombing.

Feature	IPF/UIP points toward	Alternative diagnosis points toward
Age and course	>50 years, chronic progressive dyspnea	Young age, episodic symptoms, fever, hemoptysis, acute/subacute onset
Distribution	Basal and subpleural predominance	Upper-lobe disease, peribronchovascular predominance, diffuse ground glass
Exposure	No clear antigen or occupational cause	Birds/mold/hot tub, silica/asbestos/coal, drug/radiation exposure
Systemic features	Usually absent	Raynaud, arthritis, rash, sicca, muscle weakness, uveitis, hypercalcemia

Feature	IPF/UIP points toward	Alternative diagnosis points toward
Response expectation	Antifibrotics slow decline	Inflammatory or exposure-driven disease may improve with antigen removal or immunosuppression

6. Connective tissue disease associated ILD

Connective tissue disease associated ILD is important because lung disease can be the first clue to a systemic autoimmune disorder. Systemic sclerosis can produce ILD and pulmonary arterial hypertension; rheumatoid arthritis can produce UIP-like fibrosis, pleural disease, nodules, and airway disease; inflammatory myopathies can produce rapidly progressive ILD; Sjogren disease can produce airway and interstitial patterns; lupus can cause pleuritis, pneumonitis, pulmonary hemorrhage, or thromboembolic disease. The key is not memorizing one lung pattern per rheumatologic disease, but recognizing that dyspnea plus systemic immune clues deserves a combined pulmonary-rheumatologic evaluation.

Nonspecific interstitial pneumonia is a common CTD-ILD pattern and often has more ground-glass opacity and a more uniform distribution than UIP. NSIP may be more treatment-responsive than established UIP, especially when active inflammation is present. Treatment depends on the underlying disease, pattern, severity, and pace; options may include mycophenolate, rituximab, cyclophosphamide, glucocorticoids for selected inflammatory presentations, antifibrotic therapy for progressive fibrotic phenotypes, and pulmonary hypertension-specific evaluation when the DLCO is disproportionately low.

Graph relationship

Systemic sclerosis connects to two high-yield pulmonary nodes: **ILD** and **pulmonary arterial hypertension**. A falling FVC suggests worsening restriction; a disproportionately low DLCO, exertional syncope, loud P2, or right-heart strain should activate the pulmonary hypertension pathway.

7. Hypersensitivity pneumonitis

Hypersensitivity pneumonitis is an immune-mediated lung disease triggered by inhaled antigens. The disease can be inflammatory, fibrotic, or mixed. The classic teaching examples are farmer's lung from moldy hay and bird fancier's lung from avian proteins, but real-world triggers also include feather bedding, humidifiers, hot tubs, water-damaged buildings, compost, metalworking fluids, and workplace bioaerosols. A patient may not volunteer these exposures unless asked in ordinary language: Do you keep birds? Do you sleep with feather pillows? Is there visible mold? Do symptoms improve away from home or work?

Pathologically, HP involves immune activation around small airways and alveoli. Acute or subacute HP can cause cough, dyspnea, fever, malaise, and diffuse ground-glass or centrilobular nodules. Chronic HP can progress to fibrosis and may mimic IPF. HRCT clues include mosaic attenuation, air trapping on expiratory imaging, centrilobular nodules, and a distribution that does not perfectly match classic basal subpleural UIP. Bronchoalveolar lavage lymphocytosis supports HP in the right clinical context, but no single test is perfect. Diagnosis is probabilistic and often multidisciplinary.

Treatment begins with antigen identification and avoidance. This is not always simple; removing a bird or remediating mold may be emotionally, financially, or occupationally difficult. Corticosteroids can improve inflammatory symptoms and physiology in selected cases, but they do not substitute for removing the exposure. Fibrotic HP may continue to progress even after exposure reduction and may require management similar to progressive pulmonary fibrosis.

8. Occupational and environmental lung disease

Occupational lung disease is a place where the history can outperform a broad laboratory panel. The same radiographic abnormality means different things in a stone countertop cutter, shipyard worker, coal miner, aerospace machinist, bird owner, or patient taking amiodarone. The exam question often provides the job title because the job is the mechanism. The clinical encounter requires the same discipline: ask exactly what material was cut, blasted, mined, installed, repaired, burned, or aerosolized; ask about respirator use; ask about years of exposure and latency.

Exposure-to-pattern memory map

Agent	Where to ask	Pattern	High-yield link
Asbestos	shipyard, insulation, brake work	lower-zone fibrosis, pleural plaques	lung cancer, mesothelioma
Silica	sandblasting, mining, stone cutting	upper nodules, eggshell nodes	TB risk
Coal dust	coal mining	upper nodules; massive fibrosis	COPD-like airflow disease
Beryllium	aerospace, electronics, nuclear	sarcoid-like granulomas	BeLPT confirms sensitization
Organic antigens	birds, moldy hay, hot tubs	HP: mosaic attenuation, air trapping	antigen avoidance

Asbestos-related lung disease

Asbestos exposure is classically associated with shipbuilding, insulation, demolition, construction, brake work, and older building materials. Asbestosis is a fibrotic interstitial lung disease that usually has a long latency, often measured in decades. Imaging may show lower-lobe predominant fibrosis and pleural plaques. Pleural plaques are markers of exposure and are often asymptomatic, but they are high-yield because they point toward asbestos as the exposure node. Asbestos increases risk of lung cancer and mesothelioma; cigarette smoking synergistically increases lung cancer risk in asbestos-exposed patients, although smoking is not the primary cause of mesothelioma.

Silicosis

Silicosis results from inhalation of respirable crystalline silica. High-risk settings include mining, sandblasting, foundry work, tunneling, quarrying, and engineered stone countertop cutting. Classic chronic silicosis produces upper-lobe nodular disease and may produce eggshell calcification of hilar lymph nodes. Silica impairs macrophage function and increases risk for tuberculosis. Rapidly progressive silicosis can occur after intense exposure over shorter periods, which matters for modern countertop fabrication outbreaks.

Coal workers' pneumoconiosis and progressive massive fibrosis

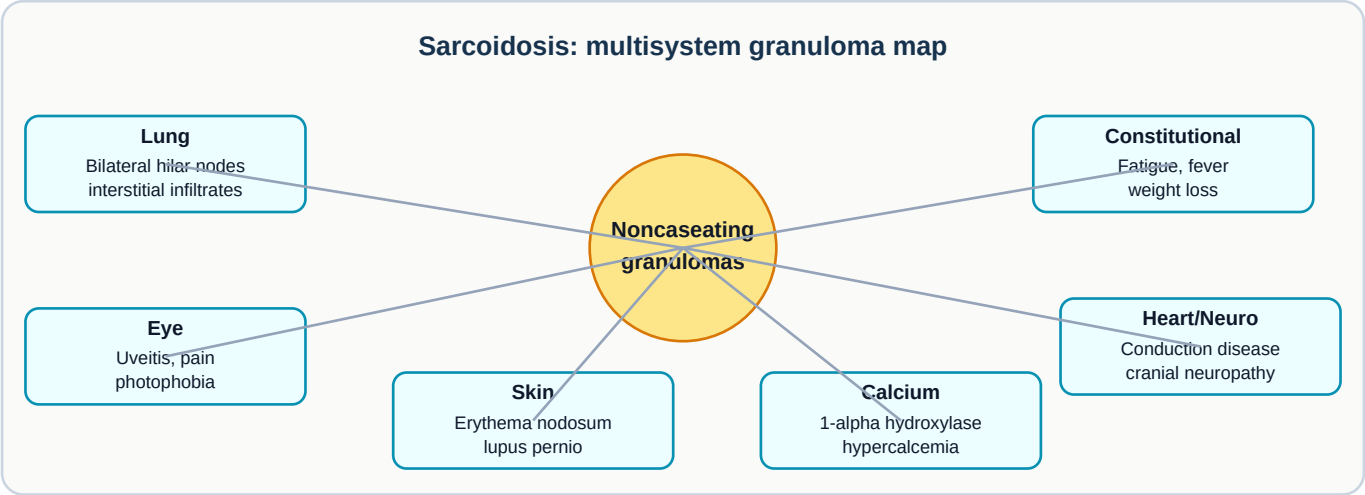
Coal mine dust exposure can produce simple coal workers' pneumoconiosis with small upper-lobe nodules and may progress to progressive massive fibrosis, in which nodules coalesce into large fibrotic masses. Patients can develop dyspnea, reduced exercise tolerance, chronic bronchitis, emphysema-like physiology, or mixed patterns. The key relationship is not just coal dust to nodules; coal dust can connect to COPD-like airflow limitation and to disabling fibrotic disease.

Beryllium disease

Chronic beryllium disease can resemble sarcoidosis because both produce granulomatous lung disease and hilar adenopathy. Exposure settings include aerospace, nuclear, electronics, ceramics, and metal machining. The beryllium lymphocyte proliferation test can help identify sensitization. The exam clue is a sarcoid-like syndrome in a patient with a beryllium exposure history; the clinical clue is any granulomatous disease in a person working with specialty metals or aerospace components.

9. Sarcoidosis

Sarcoidosis is a multisystem granulomatous disease that most commonly affects the lungs and intrathoracic lymph nodes. The diagnosis rests on three pillars: a compatible clinical-radiographic presentation, histologic evidence of nonnecrotizing granulomatous inflammation when tissue is required, and exclusion of alternative causes such as tuberculosis, fungal infection, malignancy, beryllium disease, drug reactions, and other granulomatous disorders. The word noncaseating is high-yield, but it should not make the diagnosis by itself; granulomas are a pattern, not a final answer.



Pulmonary sarcoidosis may be asymptomatic or may cause cough, dyspnea, chest discomfort, fatigue, fever, or weight loss. Chest imaging may show bilateral hilar lymphadenopathy, parenchymal infiltrates, or fibrotic disease. Extrapulmonary clues are often decisive: uveitis, erythema nodosum, lupus pernio, peripheral lymphadenopathy, hypercalcemia, kidney stones, facial nerve palsy, aseptic meningitis, cardiomyopathy, ventricular arrhythmias, or atrioventricular block. Hypercalcemia occurs because activated macrophages within granulomas can increase extrarenal 1-alpha hydroxylase activity, raising calcitriol and intestinal calcium absorption.

Not every patient with sarcoidosis needs immediate systemic therapy. Many patients with mild, asymptomatic intrathoracic disease can be observed with monitoring. Treatment is generally favored when there is significant symptomatic pulmonary disease, progressive lung function decline, or organ-threatening involvement such as ocular, cardiac, neurologic, severe cutaneous, renal, or severe hypercalcemic disease. Glucocorticoids are common first-line therapy for clinically important disease; steroid-sparing agents may be needed for chronic or relapsing disease.

Sarcoid feature	Why it matters	Mimics to exclude
Bilateral hilar lymphadenopathy	Classic thoracic clue, especially in a compatible systemic syndrome.	Lymphoma, TB, fungal infection, metastatic disease, beryllium disease.
Nonnecrotizing granulomas	Supports diagnosis when clinical context fits.	Infection, malignancy-associated granulomas, drug reactions, occupational granulomatous disease.
Hypercalcemia or kidney stones	Granuloma macrophage vitamin D activation creates calcium node.	Primary hyperparathyroidism, malignancy, vitamin D excess.
Conduction disease or syncope	Cardiac sarcoidosis can be life-threatening.	Ischemia, myocarditis, inherited arrhythmia, medication effect.

10. Treatment principles across ILD

Treatment is diagnosis-specific, but the management framework is consistent. First, remove or stop the injurious exposure when one exists: antigen avoidance in HP, dust control and exposure cessation in occupational disease, discontinuation of a culprit drug, smoking cessation, and workplace protection. Second, decide whether the dominant biology is inflammatory, fibrotic, granulomatous, or mixed. Inflammatory disease may respond to immunosuppression; fixed honeycombing usually does not reverse. Third, measure trajectory rather than relying on a single CT. Symptoms, FVC, DLCO, oxygen needs, exercise capacity, and imaging over time determine whether the disease is stable, improving, or progressive.

Supportive care is not secondary care. Oxygen should be assessed at rest and with exertion because exertional desaturation can be the earliest sign of clinically important gas-exchange failure. Pulmonary rehabilitation improves functional status and helps patients manage dyspnea. Vaccination reduces preventable respiratory infections. Comorbid pulmonary hypertension, sleep apnea, GERD, coronary disease, anemia, depression, and deconditioning can all worsen dyspnea and should not be ignored. Advanced fibrotic disease should prompt early discussion of lung transplant eligibility and goals of care rather than waiting until the patient is in crisis.

Common pitfall

Do not treat every diffuse parenchymal lung disease with prednisone by reflex. Steroids may help sarcoidosis, organizing pneumonia, inflammatory CTD-ILD, and some HP presentations, but IPF is a fibrotic disease where chronic immunosuppression is not the standard disease-modifying approach. Match the therapy to the biology.

11. Complications and red flags

- Pulmonary hypertension: suspect when dyspnea is disproportionate, DLCO is very low compared with FVC, oxygen need rises, P2 is loud, edema or syncope develops, or CT/echo suggests right-heart strain.
- Acute exacerbation of ILD: new bilateral ground-glass opacities and acute respiratory worsening without another clear cause; infection, pulmonary edema, PE, and drug injury must be excluded.
- Lung cancer: risk is increased in several fibrotic and occupational lung diseases; new focal opacity, hemoptysis, weight loss, or asymmetric findings require evaluation.
- Infection: immunosuppression for ILD or sarcoidosis increases risk; distinguish flare from pneumonia before escalating steroids.
- Respiratory failure: ILD patients can decompensate with small insults because diffusion reserve is limited and stiff lungs are difficult to ventilate.

12. Differentials and mimics

ILD is frequently confused with heart failure because both can cause exertional dyspnea, crackles, hypoxemia, and abnormal imaging. Clues for heart failure include orthopnea, edema, elevated natriuretic peptides, cardiomegaly, pleural effusions, and rapid response to diuresis. Clues for ILD include chronic dry cough, clubbing, persistent exertional desaturation, reduced TLC, reduced DLCO, and HRCT reticulation or honeycombing. The two can coexist, especially in older patients.

Infection is another mimic. Diffuse ground-glass opacity can represent viral pneumonia, *Pneumocystis pneumonia*, pulmonary edema, hemorrhage, drug toxicity, HP, sarcoidosis, or an acute ILD exacerbation. Timing, immune status, fever, leukocytosis, cultures, bronchoscopy, exposure history, and response to treatment help separate these. Malignancy can mimic ILD through lymphangitic spread, drug-induced pneumonitis from cancer therapy, radiation fibrosis, or postobstructive changes.

Finding	Think ILD	Think mimic
Crackles	Fine, late inspiratory, bibasilar, persistent	Wet crackles with volume overload may improve after diuresis
Wheeze	Can occur in sarcoid/HP/airway-centered disease	Asthma, COPD, heart failure, central airway lesion
Ground glass	Inflammatory ILD, HP, NSIP, acute exacerbation	Edema, infection, hemorrhage, drug toxicity
Adenopathy	Sarcoidosis, beryllium disease	Lymphoma, TB, fungal disease, metastases
Low DLCO	Parenchymal fibrosis or pulmonary vascular involvement	Emphysema, anemia, pulmonary hypertension, chronic PE

13. High-yield exam clues

Clue	Most likely association	Reasoning link
Older patient, dry cough, bibasilar Velcro crackles, honeycombing	Idiopathic pulmonary fibrosis	UIP pattern plus chronic progression.
Bird exposure, mosaic attenuation, air trapping, BAL lymphocytosis	Hypersensitivity pneumonitis	Inhaled antigen drives small-airway/alveolar immune response.
Shipyard/insulation exposure, pleural plaques, lower-lobe fibrosis	Asbestos-related disease	Pleural plaques are exposure markers; fibrosis often basal.

Clue	Most likely association	Reasoning link
Sandblasting or stone cutting, upper-lobe nodules, eggshell calcification	Silicosis	Respirable crystalline silica exposure; increased TB risk.
Bilateral hilar lymphadenopathy, uveitis, erythema nodosum	Sarcoidosis	Systemic nonnecrotizing granulomatous disease.
Sarcoid-like disease in aerospace/electronics worker	Chronic beryllium disease	Occupational granulomatous mimic of sarcoidosis.
Raynaud plus low DLCO and dyspnea	Systemic sclerosis ILD or pulmonary hypertension	Scleroderma connects to both interstitial and vascular pulmonary disease.

14. Practice questions

Question 1. A 68-year-old former smoker has 18 months of progressive exertional dyspnea and dry cough. Exam shows bibasilar fine inspiratory crackles. Spirometry shows reduced FVC with preserved FEV1/FVC; DLCO is reduced. HRCT shows basal subpleural reticulation and honeycombing. What is the most likely pattern?

Answer

Usual interstitial pneumonia pattern consistent with idiopathic pulmonary fibrosis after exclusion of secondary causes. The preserved ratio with low FVC suggests restriction, and honeycombing supports established fibrosis.

Question 2. A 45-year-old develops cough and dyspnea that worsen in winter. HRCT shows centrilobular nodules, mosaic attenuation, and air trapping. The patient keeps several birds and uses feather bedding. What management step matters most?

Answer

Identify and remove the antigen exposure. Corticosteroids may help inflammatory disease, but continued exposure can drive recurrence or fibrosis.

Question 3. A construction worker with prior insulation exposure has pleural plaques and lower-lobe fibrosis. Which malignancies are associated with the exposure?

Answer

Asbestos exposure increases risk for lung cancer and mesothelioma. Smoking synergistically increases lung cancer risk in asbestos-exposed patients.

Question 4. A stone countertop cutter develops upper-lobe nodules and hilar lymph node calcification. What infection risk should be considered?

Answer

Tuberculosis risk is increased in silicosis because silica impairs macrophage function.

Question 5. A young adult has bilateral hilar lymphadenopathy, ankle arthritis, erythema nodosum, and eye pain. Calcium is mildly elevated. What mechanism explains the calcium abnormality?

Answer

Activated macrophages in granulomas can increase extrarenal 1-alpha hydroxylase activity, raising calcitriol and calcium absorption.

15. Graph-RAG extraction map

The following concept map is designed for automated extraction. Nodes should include disease entities, physiologic patterns, exposures, imaging findings, symptoms, diagnostic tests, treatments, and complications. Edges should encode causal, diagnostic, associated-with, treats, excludes, and complication relationships.

Source node	Edge	Target node	Rationale
Interstitial lung disease	causes	Restrictive physiology	Interstitial inflammation/fibrosis reduces lung compliance and total lung capacity.
Restrictive physiology	supported by	Low TLC	TLC confirms restriction when spirometry suggests it.
ILD	causes	Reduced DLCO	Thickened or destroyed alveolar-capillary surface impairs diffusion.
IPF	associated with	UIP pattern	UIP is the classic imaging/pathologic pattern in IPF.
UIP pattern	imaging feature	Honeycombing	Honeycombing indicates established fibrotic remodeling.
Bird or mold exposure	triggers	Hypersensitivity pneumonitis	Inhaled organic antigens drive immune-mediated lung injury.
Asbestos	associated with	Pleural plaques	Pleural plaques are a common marker of asbestos exposure.
Silica	increases risk of	Tuberculosis	Silica impairs macrophage function.
Sarcoidosis	histology	Nonnecrotizing granulomas	Granulomas are central to diagnosis after excluding mimics.
Sarcoid granulomas	cause	Hypercalcemia	Macrophage 1-alpha hydroxylase increases calcitriol production.
Fibrotic ILD	complication	Pulmonary hypertension	Chronic hypoxemia and vascular remodeling strain the right ventricle.

Selected source anchors used for accuracy checking

This original chapter was cross-checked against contemporary pulmonary guidance and occupational health references, including ATS/ERS/JRS/ALAT guidance on IPF and progressive pulmonary fibrosis, ATS/JRS/ALAT guidance on hypersensitivity pneumonitis diagnosis, ATS sarcoidosis diagnosis guidance, and CDC/NIOSH occupational lung disease resources. These sources were used for medical accuracy, not for wording or figures.

1. ATS/ERS/JRS/ALAT. Idiopathic Pulmonary Fibrosis (an Update) and Progressive Pulmonary Fibrosis in Adults guideline resources. 2022.
2. ATS/JRS/ALAT. Diagnosis of Hypersensitivity Pneumonitis in Adults. Official clinical practice guideline. 2020.
3. American Thoracic Society. Diagnosis and Detection of Sarcoidosis. Official clinical practice guideline. 2020.
4. CDC/NIOSH. Pneumoconioses and respiratory health resources for occupational lung disease.